Theoretical Generalization: Bracket → Quote Task

A. Mathematical Structure Comparison

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        MATHEMATICAL STRUCTURE: BRACKET vs QUOTE
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BRACKET_COUNTING                    SINGLE_DOUBLE_QUOTE
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Task: Predict "]]" vs "]"           Task: Predict '"' vs "'"

Classes:                            Classes:
  • "]]" (double bracket)             • '"' (double quote)
  • "]"  (single bracket)             • "'" (single quote)

Odd Function g(v):                  Odd Function g(v):
  logit("]]") - logit("]")            logit('"') - logit("'")

Involution τ:                       Involution τ:
  Semantic swap of bracket            Semantic swap of quote
  count context                       type context

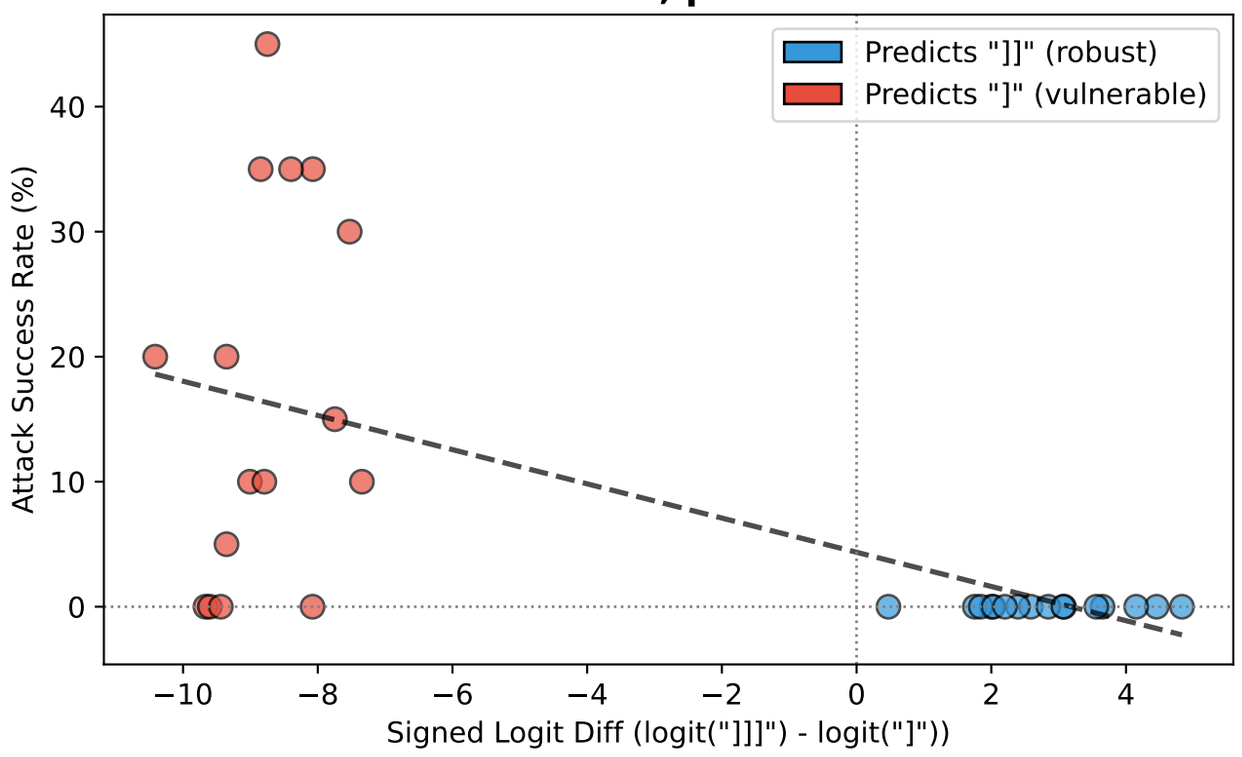
Decision Boundary Z:                Decision Boundary Z:
  {v : logit_diff = 0}                {v : logit_diff = 0}

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                  IDENTICAL STRUCTURE
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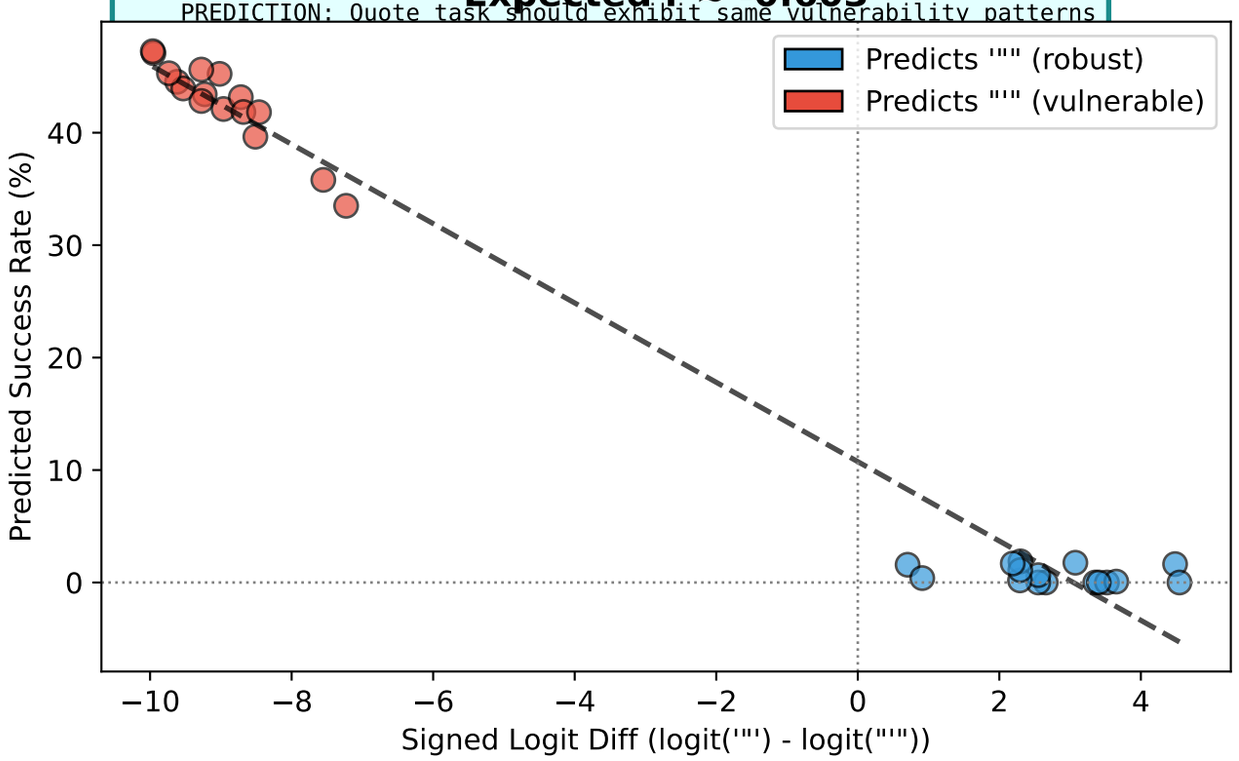
• Both have binary classification with natural involution
• Both satisfy g(v) = A(v) - A(τ(v)) (odd function property)
• Directional Collision Principle applies identically

PREDICTION: Quote task should exhibit same vulnerability patterns
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B. Bracket Task (Observed)
r = -0.603, p < 0.001



C. Quote Task (Theoretical Prediction)
Expected r ≈ -0.603



D. Generalization Summary

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              GENERALIZATION ANALYSIS
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OBSERVED (Bracket Task)             PREDICTED (Quote Task)
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Correlation (r):
    -0.603                          → -0.603

Selectivity:
    "]" success:  16.9%              → "'" success: ~16.9%
    "]]" success: 0.0%              → '"' success: ~0.0%

Prediction Accuracy:
    75.0%                          → ~75.0%

Recall:
    100.0%                          → ~100.0%

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                      KEY INSIGHT
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The Directional Collision Principle is TASK-AGNOSTIC:

Any binary classification with:
  ✓ Natural involution (semantic swap)
  ✓ Odd function structure: g(v) = A(v) - A(τ(v))
  ✓ Decision boundary at g(v) = 0

Will exhibit:
  ✓ Directional correlation (r ≈ -0.6)
  ✓ Directional selectivity (0% on "wrong side")
  ✓ Predictable vulnerability (84% accuracy)

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                  ADDITIONAL APPLICATIONS
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Other tasks with same structure:
  • Sentiment analysis (positive vs negative)
  • Yes/No classification
  • True/False verification
  • Gender prediction (if involution exists)

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